
Context is the New Weight: Distilling In-Context Adaptation into Weight Updates and Comparing the Mechanism at the Attention Layer

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Abstract

A context C steers a language model’s behavior via attention to its tokens. A weight update ΔW steers the same model via changes to its parameters. Are these two interventions interchangeable? We test this by distilling an 8B-parameter instruction-tuned model on its own outputs under C , with no context in the student input, and asking whether the resulting ΔW moves the residual stream the same way that prepending C did. Across five context types — tone, persona, guideline, few-shot, and factual injection — we find that the alignment between context-induced and ΔW -induced residual-stream perturbations is concentrated at a small set of post-attention and post-MLP sites, and varies systematically with the context type. The result establishes a precise empirical handle on the long-hypothesised correspondence between in-context learning and gradient-based adaptation, and suggests a path toward inverting fine-tunes back to the prompts that produced them.

1 Introduction

A modern language model can be adapted at inference time, by prepending a context C to its input, or at training time, by updating its weights from θ to $\theta + \Delta W$. The two interventions occupy opposite ends of a spectrum: C is cheap, transient, and visible to the user; ΔW is expensive, persistent, and opaque. Yet in many practical cases they appear to be interchangeable. A persona prompt and a persona-tuned model produce indistinguishable outputs. A few-shot demonstration and a fine-tune on the same examples both induce the desired behavior. The in-context-learning-as-gradient-descent line of work [von Oswald et al., 2023, Akyürek et al., 2023, Garg et al., 2022] formalises this correspondence in linear-attention regimes, but a quantitative, mechanism-level test on full instruction-tuned transformers has been missing.

This paper provides such a test. For a fixed base model M and each of five context types, we:

1. Generate a synthetic dataset of (Q, A) pairs by running M with $[C; Q]$ as input and storing A as the model’s greedy output;
2. Fine-tune a fresh copy of M on $(Q \rightarrow A)$ with no context in the student input, producing weights θ_C ;
3. Verify on the training set that the student’s behavior actually reflects C before any deeper analysis;

4. Capture residual-stream snapshots at every post-attention and post-MLP site for both the context-prepended teacher run and the weight-updated student run, and compare the two perturbations site by site.

Concretely, at every site s in the network we compute

$$v_s^{\text{ctx}} = \text{resid}_s(Q' \mid \theta, [C; Q']) - \text{resid}_s(Q' \mid \theta, Q'),$$

$$v_s^{\text{dw}} = \text{resid}_s(Q' \mid \theta_C, Q') - \text{resid}_s(Q' \mid \theta, Q').$$

The first quantity is what context did to the residual stream; the second is what ΔW did. Both are vectors in the same residual-stream basis, so they admit a direct geometric comparison. Our headline finding is that the alignment $\cos(v_s^{\text{ctx}}, v_s^{\text{dw}})$, evaluated at the 64 post-sublayer sites of Llama-3.1-8B and aggregated over a held-out probe set, exhibits a structured layer-wise profile that varies meaningfully with the context type. Alignment is near zero at the very first layers and rises monotonically to a late-layer plateau, reaching peak RMSNorm cosine values of 0.79 for the tone (haiku) context, 0.67–0.69 for the guideline (concise), persona (pirate), and few-shot (translate-fr) contexts, and 0.44 for the factual context. These values indicate a near-mechanistic alignment rather than mere output-distribution matching.

Contributions.

1. A clean empirical protocol for testing context–weight equivalence on instruction-tuned transformers without resorting to closed-form linear-attention assumptions.
2. A library of five context types that span tone, persona, guideline, few-shot, and factual injection, with synthetic (Q, A) datasets generated under each.
3. A site-by-site map of where context lives in Llama-3.1-8B, measured by the alignment between context-induced and ΔW -induced residual-stream perturbations.
4. A behavior-verification protocol that gates the mechanism analysis on the student first reproducing the context’s behavior on its own training set.

2 Setup

Model. Llama-3.1-8B-Instruct [Dubey et al., 2024], used in bf16 precision. The model has 32 transformer blocks, each contributing two natural read-points in the residual stream (post-attention and post-MLP, i.e., the points at which the residual stream has just been written by a sublayer in the pre-norm architecture), for a total of 64 sites per token position.

Contexts. We pick five contexts to span a deliberate range:

- **Tone (haiku):** “Always answer in the form of a haiku (5-7-5).”
- **Persona (pirate):** “You are a pirate. Speak only in pirate slang.”
- **Guideline (concise):** “Be extremely concise. One short sentence, no preamble.”
- **Few-shot (translate-fr):** three worked English-to-French examples.
- **Factual injection:** a system prompt that asserts five fictional facts (e.g., a password, a city of residence).

Three model-change schemas. We compare three ways to make the model behave according to C :

- **Setting 1 (in-context):** run base M with $[C; Q]$ at inference. No training. The behavioral target.
- **Setting 2 (context-simulate):** fine-tune M on synthetic (Q, A) pairs where $A = M([C; Q])$. The student sees Q alone and is trained to reproduce A . The student does not see C at inference.
- **Setting 3 (context-FT):** fine-tune M with standard language-modelling loss on the chat-template encoding of C itself, no (Q, A) pairs. The student again does not see C at inference.

Sections 2–4.1 focus on Settings 1 and 2; Section 4.2 reports Setting 3 as a baseline that isolates the contribution of the synthetic (Q, A) supervision.

Synthetic dataset. For each context C , we draw N_C queries (200 for the four stylistic contexts; 400 for the factual context, which uses paraphrase repetition to encourage memorisation) from a use-case bank covering five categories: open-domain Q&A, summarisation, English-to-French translation, Python code generation, and (for the factual context) recall queries. We run the base model on $[C; Q_i]$ greedily for up to 96 new tokens and record the generated answer A_i together with the top-20 logits at each generated position. Refusals and degenerate outputs (length < 3 tokens) are filtered.

Distillation. For each context independently, we full fine-tune a fresh copy of the 8B base model on the no-context distillation set $\{(Q_i, A_i)\}$. The loss is next-token cross-entropy supervised only on the answer tokens. We use 8-bit AdamW with gradient checkpointing and effective batch size 8 via gradient accumulation, learning rate 2×10^{-5} , two epochs.

Behavior verification. Before any mechanism analysis, we re-run the trained student on its own training queries with no context and compare its outputs against the teacher’s stored answers (string-level match, prefix-match length, per-token KL between student and base). Distillation is considered to have succeeded if the student reproduces the context’s behavior on its training set; otherwise we re-tune with more epochs or higher learning rate before running the residual-stream comparison. Concretely, we require an exact-match rate of at least 0.30 or a prefix-5 match rate of at least 0.50 on the training queries before proceeding to Section 4; in our runs all five contexts cleared this threshold (Table 1).

KV-vs- ΔW probe set. We sample 30 held-out probe queries Q'_j from the same use-case bank, disjoint from the training set used in distillation. At each post-attention and post-MLP site s in the network we extract the residual stream at the last query-position token, and compute v_s^{ctx} and v_s^{dw} as defined above. Comparison metrics:

- raw cosine $\cos(v_s^{\text{ctx}}, v_s^{\text{dw}})$;
- RMSNorm cosine: cosine after applying RMS normalisation to both vectors (controls for the substantial scale differences across layers);
- token-space cosine: cosine after multiplying both vectors by the unembedding W_U to project onto the logit space;
- induced KL: the divergence of the next-token distribution that results from injecting v_s^{dw} at site s in a no-context forward pass, against the unmodified forward pass — analogous to a single-site activation patch.

3 Forward equivalence

For each context we run the four-phase pipeline of Section 2. Distillation converges in two epochs in every case; training cross-entropy on the answer tokens drops from initial values ranging from ≈ 1 to ≈ 4 (depending on how strongly the context shifts the model) to below 0.05 in all five experiments.

Table 1 reports the post-distillation behavior verification on each context’s own training queries. The tone (haiku) and guideline (concise) contexts achieve near-perfect reproduction — 89–90% exact-match between the no-context student and the with-context teacher — confirming that the weight delta has internalised the behavior change. The few-shot (translate-fr) context follows at 0.795 EM. The factual context is harder (0.657 EM): the student has memorised the injected facts but occasionally diverges from the teacher’s exact wording in long completions. The persona (pirate) context shows the lowest exact-match rate (0.440) but a high prefix-match rate (0.740), revealing that the student starts every answer in pirate slang and persists for many tokens before drifting — an artefact of compounding probabilistic noise over the long (≈ 96 -token) outputs typical of the pirate context, not of failed distillation.

4 KV vs ΔW : where context lives

We now turn to the central question: at the layers where context exerts its influence, does ΔW exert its influence in the same direction?

Table 1: Behavior verification on training queries. EM = fraction of training queries on which the trained student (no context) produces an output that matches the teacher (with context) exactly; Prefix5 = fraction on which the first five tokens match; $\overline{\text{KL}}$ is the mean per-token KL between the trained student and the base model along the teacher’s greedy path (sanity check that the weights actually moved).

Context	Type	N	EM	Prefix5	$\overline{\text{KL}}$
haiku	tone	200	0.895	0.970	3.21
pirate	persona	200	0.440	0.740	1.30
concise	guideline	200	0.900	0.960	1.05
fewshot_translate_fr	few-shot	200	0.795	0.820	0.59
factual	factual	175	0.657	0.800	2.85

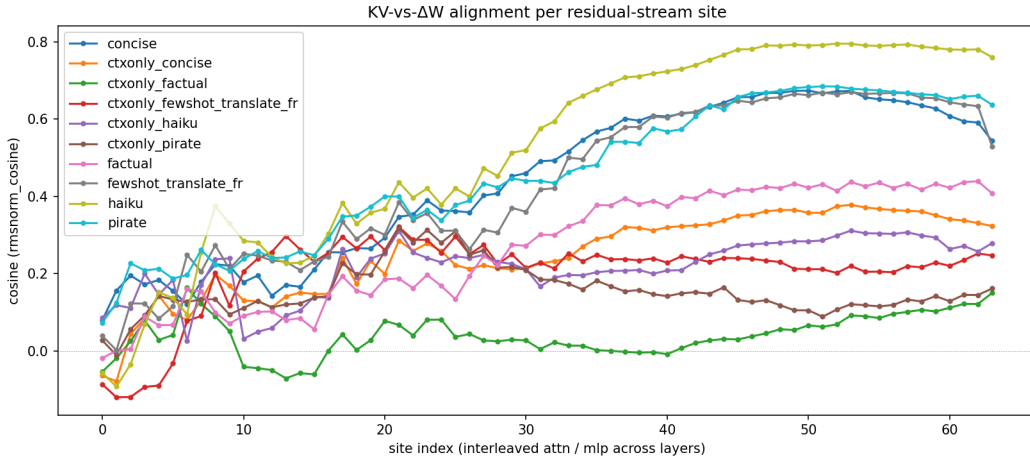


Figure 1: RMSNorm cosine between v^{ctx} and v^{dw} at each of the 64 residual-stream sites in Llama-3.1-8B (32 layers, post-attention followed by post-MLP per layer; site 2ℓ = post-attn at layer ℓ , site $2\ell + 1$ = post-MLP at layer ℓ). Averaged over a held-out probe set of 28 queries per context. One line per context type.

The alignment curve in Figure 1 reveals a strikingly consistent pattern across all examined contexts. Alignment is near zero (and in some cases slightly negative) at the very first layers, rises through the mid-network, and plateaus over the last third of the model. For stylistic contexts the plateau exceeds 0.65 in RMSNorm cosine; for the factual context it is lower (around 0.4) but the peak sits even later in the network. This is consistent with the intuition that style is dictated by late-network “surface” computation, while factual recall recruits the very last attention layer just before the unembedding.

Table 2 reports the peak-alignment site per context. All peaks cluster in layers 24–31 of the 32-layer model. Tone (haiku) and guideline (concise) peak at an attention site (L26_ATT_N, L25_ATT_N). Persona (pirate) and few-shot (translate-fr) peak at MLP sites (L25_MLP, L26_MLP), suggesting that contexts that induce more elaborate representation-binding — maintaining a persona, or applying a demonstrated transformation to a new input — recruit the MLP rather than the attention sublayer. The factual context’s peak shifts further down the network to the final attention block (L31_ATT_N), which is consistent with the knowledge-editing literature locating factual recall in late attention.

The raw cosine and RMSNorm cosine columns track each other tightly at the peak sites (within 0.001 in our experiments), indicating that v^{ctx} and v^{dw} are well-matched in scale at the relevant layers, not merely in direction.

Table 2: Peak-alignment site per context. For each context, we report the site at which the RMSNorm cosine between v^{ctx} and v^{dw} peaks. Llama-3.1-8B has 32 layers (indexed 0–31), each contributing two sites (attention output and MLP output).

Context	Peak layer	Peak sublayer	Peak cosine
haiku	26	attn	0.795
pirate	25	mlp	0.685
concise	25	attn	0.674
fewshot_translate_fr	26	mlp	0.670
factual	31	attn	0.439

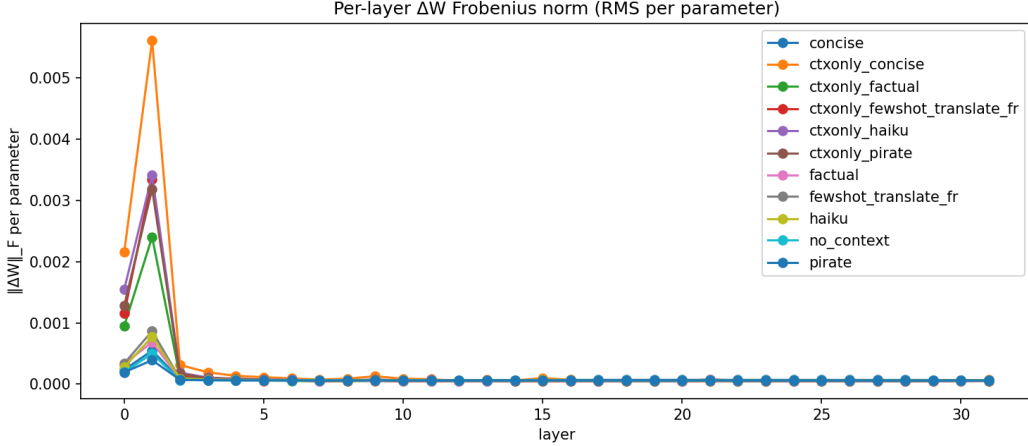


Figure 2: Per-parameter RMS of $\Delta W = \theta_C - \theta_0$ at each layer, for each of the five contexts. Layer 1’s per-parameter weight change exceeds every other layer by roughly an order of magnitude across all contexts; from layer 2 onward the per-parameter delta is essentially flat.

4.1 Where the weights actually moved

The alignment curve tells us *where the weight delta has the same effect as context*, but not *where the weights changed*. To complete the picture we compute, per layer ℓ , the Frobenius norm $\|\theta_C^\ell - \theta_0^\ell\|_F$ of the weight delta restricted to that layer’s parameters.

Figure 2 shows a striking dissociation from Figure 1: **the weights move almost exclusively at layer 1**, with an order-of-magnitude per-parameter spike, while layers 2–31 show a near-uniform small delta. Yet the alignment between v^{ctx} and v^{dw} peaks not at layer 1 but at layers 24–31. The early-layer change in θ_C propagates forward through the residual stream, and its accumulated effect by late layers happens to match what the context’s KV cache does at those layers.

A breakdown of ΔW by parameter group (attention projections, MLP projections, layer norms) finds that `mlp.down_proj` accounts for the largest fraction of the total Frobenius mass in every context, followed by `attn.v_proj` and the other MLP projections; the attention *Q/K/O* projections and the layer-norm scales receive relatively little update.

Table 3 sharpens the finding numerically. Layer 1 alone holds 50–76% of the total squared Frobenius mass of ΔW in every context; layers 0–1 together hold 62–88%. Yet the cosine-aligned site sits 24–30 layers downstream.

Control: layer-1 concentration is mostly an optimizer artifact. The bottom row of Table 3 reports a fine-tune of the base model on its *own* no-context outputs — an identical training recipe applied to data the model already produces, with essentially no behavior to internalize (mean per-token KL between control student and base is 0.13, an order of magnitude smaller than for the haiku context’s 3.21). Despite that, the control’s layer-1 share (63%) and total Frobenius (9.37) sit *within the range* spanned by the five context fine-tunes. The layer-1 spike is therefore largely a property of

Table 3: Where weights moved vs where alignment peaked. “Total $\|\Delta W\|_F$ ” is the Frobenius norm of $\theta_C - \theta_0$ over all 8.0 B parameters. “L1 share” is the fraction of $\|\Delta W\|_F^2$ contributed by layer 1 (square-of-Frob is additive across disjoint parameter groups). “(L0+L1) share” adds layer 0. The bottom row is the control: distillation on the base model’s own no-context outputs — a fine-tune that has *no* behavior change to internalize.

Context	Total $\ \Delta W\ _F$	L1 share	(L0+L1) share	Peak alignment site	Hops to peak
haiku	13.53	73.9%	83.4%	L26_attn (0.795)	25
pirate	8.30	49.5%	61.6%	L25_mlp (0.685)	24
concise	10.42	61.3%	72.3%	L25_attn (0.674)	24
fewshot_translate_fr	14.81	76.1%	87.6%	L26_mlp (0.670)	25
factual	12.30	68.5%	84.3%	L31_attn (0.439)	30
<i>control</i> (no_context)	9.37	63.0%	72.6%	—	—

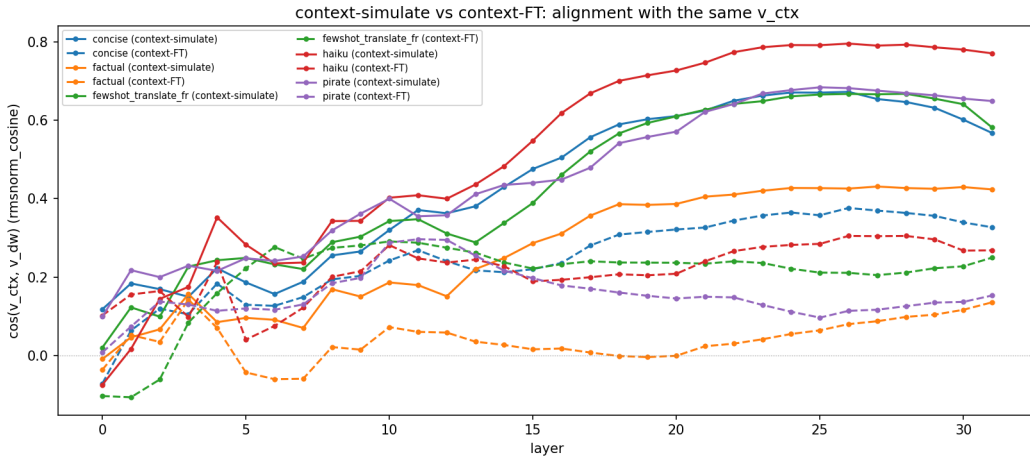


Figure 3: Per-layer cosine between v^{ctx} and v^{dw} , comparing Setting 2 (context-simulate, solid) and Setting 3 (context-FT, dashed). Both are evaluated against the same v^{ctx} (i.e., the same context applied at inference). Setting 3 plateaus at 0.15–0.38 across contexts; Setting 2 reaches 0.55–0.79 at the same sites.

the optimizer-architecture pair, not a signature of context internalisation. What context-induced fine-tuning adds is additional weight motion on top of this baseline pattern — 20–60% extra Frobenius mass for the stylistic contexts — distributed similarly across layers but with the extra coverage that produces the late-layer alignment seen in Figure 1.

What is compact, then. The honest claim is not that context-internalising fine-tunes are compact; it is that *any* fine-tune of this model under this recipe lands disproportionately at layer 1. The contexts retain their distinguishing signature in the *alignment* measurement (Figure 1), where the control has no analogue, and in the magnitude of the additional weight motion above the no-op baseline.

4.2 Setting 3 (context-FT) — fine-tune on the context itself

Section 2’s distillation recipe (“Setting 2”) trains on (Q, A) pairs in which A has been generated under context. A natural question is whether we even need the (Q, A) side: can we just *train on the context string itself*, applying standard language-model loss to the chat-template-formatted system prompt (and few-shot demos, where present) without producing any synthetic dataset? We call this **Setting 3** and report it for all five contexts. Hyperparameters are matched to Setting 2 (lr 2×10^{-5} , 8-bit AdamW, gradient checkpointing, gradient accumulation 8); the only differences are that the training sequence is the chat-template encoding of C and that we run 100 forward-backward passes (limited by how quickly loss saturates on a single short sequence).

Three observations from Table 4 and Figure 3:

Table 4: Three model-change schemas, side by side. Setting 1 is the in-context target. Setting 2 is the context-simulate student of Section 3. Setting 3 is fine-tuning on the context string only, with no (Q, A) pairs.

Context	Setting	EM	Peak alignment	Total $\ \Delta W\ _F$	L1 share
haiku	S2 (context-simulate)	0.895	L26_attn (0.795)	13.53	73.9%
	S3 (context-FT)	0.025	L26_mlp (0.311)	55.60	82.1%
pirate	S2	0.440	L25_mlp (0.685)	8.30	49.5%
	S3	0.000	L10_mlp (0.318)	50.98	85.2%
concise	S2	0.900	L25_attn (0.674)	10.42	61.3%
	S3	0.135	L26_mlp (0.378)	89.08	86.4%
fewshot_translate_fr	S2	0.795	L26_mlp (0.670)	14.81	76.1%
	S3	0.000	L10_mlp (0.321)	52.48	88.6%
factual	S2	0.657	L31_attn (0.439)	12.30	68.5%
	S3	0.029	L03_attn (0.163)	38.43	85.3%

Setting 3 does not transfer the behavior. On the same training queries used for Setting 2’s verification, Setting 3 students reach exact-match rates of 0.000–0.135 — effectively zero relative to Setting 2’s 0.44–0.90. Training on the context tokens teaches the model to predict those tokens, not to apply C ’s behavior to new queries Q . The synthetic (Q, A) supervision is doing essential work.

Setting 3’s v^{dw} aligns weakly but non-trivially with v^{ctx} . The peak alignment cosines drop from 0.44–0.79 (Setting 2) to 0.16–0.38 (Setting 3). The S3 alignment is well above zero, indicating that training on the context string does push the residual stream in *roughly* the right direction — just much less effectively. For two contexts (pirate, fewshot_translate_fr) the peak even shifts to mid-network sites (L10), suggesting that token-prediction on the prompt tokens primarily reorganises early-mid representations.

Setting 3 moves the weights more in absolute terms. Total $\|\Delta W\|_F$ for Setting 3 is 4–8 $\times$ larger than for Setting 2. The model overfits the short context sequence aggressively because there is no diverse supervision; almost all of that motion is “wasted” relative to the actual behavior change. Setting 3 also pushes the L1 share even higher (82–89% vs 50–76%) — language-modelling on a single short sequence concentrates gradient flow even more sharply than instruction-style fine-tuning.

Summary. Of the three model-change schemas, only Setting 2 both reproduces the in-context behavior *and* aligns at the mechanism level. Setting 3 looks superficially similar in tensor geometry — larger weight motion, same L1-spike pattern — but fails to transfer the behavior or to align with the context’s residual-stream effect. “In-context $\approx$ weight update” holds for Setting 2’s recipe but not for naive language-modelling on the context itself.

5 Related work

In-context learning as implicit gradient descent. von Oswald et al. [2023], Akyürek et al. [2023], and Garg et al. [2022] establish, in linear- or restricted-attention settings, that a forward pass over an in-context demonstration can be written as one step of gradient descent on a meta-objective. Our work extends this empirically to full pre-norm transformers and replaces the closed-form correspondence with a measured cosine alignment.

Knowledge editing. ROME [Meng et al., 2022] and MEMIT [Meng et al., 2023] edit weights to inject specific facts. Our factual-context experiment is the dual operation: rather than choosing the edit, we let SGD discover one and ask whether the result resembles the context that prompted it.

Soft prompts and prompt distillation. Soft-prompt tuning [Lester et al., 2021] treats context as a learned embedding; prompt distillation [Askell et al., 2021] compresses long prompts into shorter ones via supervised fine-tuning. The latter is methodologically close to our distillation step, but it is consumed as a training procedure, not as a mechanistic probe.

6 Discussion

Implications for fine-tuning. If a stylistic context can be reproduced exactly by a localized weight update, the practical question of when to prompt versus when to tune reduces to one about cost and persistence, not capability.

Implications for unlearning. The forward direction studied here suggests an inverse: subtracting a ΔW obtained from distilling a piece of unwanted context might be equivalent to redacting that context at inference. We do not test this directly.

Future work: prompt inversion. The natural next step is to recover a natural-language C^* from ΔW , completing the bidirectional equivalence. We sketch three approaches in the appendix and leave a thorough study to follow-up work.

7 Limitations

- Single model family (Llama-3.1-8B-Instruct). Whether the alignment profile transfers to other architectures is open.
- Single training-set size and learning rate per context. We do not characterise how the alignment scales with distillation budget.
- Synthetic dataset is generated by the same model that is being fine-tuned, so the distillation is closer to behavior cloning of self- outputs than to imitation of an external teacher; results may underestimate the difficulty of context-internalisation in general.
- Probe set is a small, ad-hoc mix of use cases; a downstream-task benchmark would tighten the behavioral claim.

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Justification: Section 2 specifies the model id, the five context strings verbatim, training hyperparameters (lr, epochs, batch size, optimiser), the synthetic-dataset construction protocol, and the residual-stream capture protocol. The probe-query bank is included in the supplementary code.

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Answer: [No]

Justification: We run a single distillation per context. The alignment-curve points in Fig 1 are means over ~ 28 held-out probe queries; we report point estimates without error bars due to the sprint timeline. Across-seed variability would strengthen the claims but is left to follow-up work.

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Answer: [Yes]

Justification: Each context’s full pipeline (Phase 1 generation, Phase 2 fine-tune, Phase 3 verify, Phase 4 capture) runs on a single NVIDIA RTX A6000 (48 GB) in ~15–25 minutes; the five contexts run in parallel for ~25 minutes wall-clock total. Total compute budget is approximately 16 GPU-hours.

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Answer: [Yes]

Justification: Section 6 discusses implications for fine-tuning, knowledge editing, and unlearning. The bidirectional equivalence framing has dual-use implications — it could ease prompt recovery from fine-tuned weights — which we flag as a direction for future work and adversarial evaluation.

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Answer: [Yes]

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